

Concise Update: Baseline MLP $\rightarrow$ MLP Run 005

Hexapod RL Project

February 13, 2026

Goal

Improve realism of servo-driven locomotion while keeping forward movement:

- less high-frequency wobble,
- clearer lift–swing–land leg behavior,
- stable movement on path.

What We Changed (vs baseline `mlp_baseline`)

Timing and smoothness

- Phase/imitation frequency: 0.9 Hz $\rightarrow$ 0.45 Hz
- Action smoothing enabled: `action_smooth_alpha` = 0.6
- Action-rate cap: `action_rate_limit` = 0.15

Contact-aware gait shaping

- Contact added to observation (`obs_include_contact` = true)
- Stance penalties to reduce motion while foot is planted: `stance_stability` = 0.2, `stance_action` = 0.35

Servo-friendly regularization

- `smoothness` = 2.5, `jerk` = 0.35, `joint_velocity` = 0.04

Step geometry and movement target

- Larger leg motion range: `abd`: 35 $\rightarrow$ 42 deg, `flex`: 60 $\rightarrow$ 72 deg, `max flex`: 70 $\rightarrow$ 88 deg
- Target-speed shaping replaced raw velocity push: `target_forward_velocity` = 0.09, `target_velocity_weight` = 28.0

Results Snapshot (Evaluation)

Metric	Baseline MLP	MLP Run 005
Mean reward	2817.78	21962.45
Reward std. dev.	3208.06	71.16
Mean forward velocity (m/s)	0.01266	0.01881
Max forward velocity (m/s)	1.0795	0.1750
Mean energy proxy ($\sum a^2$)	3.8585	1.0746

Interpretation:

- Run 005 is significantly more stable (much lower reward variance).
- Average forward speed improved.
- Extreme spikes dropped; control became smoother and more servo-plausible.
- Energy proxy decreased, consistent with less jitter.

Visualization Upgrade

Evaluation videos now include per-frame overlays (step, (x, y) , forward distance, velocity), and each episode saves a trajectory plot (`_trajectory.png`) for quick movement diagnostics relative to the environment.

Bottom Line

Compared with baseline MLP, `mlp_run_005` delivers cleaner, more insect-like, servo-realistic locomotion while preserving useful forward movement.